Aptech Computer Education Examination

Xcode and iOS Application Development using Swift - Comprehensive Exam Sample

Exam Code: MOD-IOS-SWIFT-401 | Total Questions: 44 | Total Marks: 44.00 | Passing Marks: 17.00

Question 1: Which one of the following options will you use to manage class name for an object in Xcode?

a) Quick Help
b) Attributes inspector
c) **Identity inspector**
d) Connections inspector

Reference / Note: The Identity Inspector in Xcode allows setting custom classes, runtime attributes, and accessibility identifiers for UI objects.

Question 2: Which one of the following options will you use to define actions and outlets for an interface object?

a) File inspector
b) Identity inspector
c) Bindings inspector
d) **Connections inspector**

Reference / Note: The Connections Inspector handles connections (outlets and actions) between UI objects and source code.

Question 3: Which of the following is used to automate memory management for Objective-C objects?

a) nib file
b) Storyboards
c) Unit tests
d) **ARC (Automatic Reference Counting)**

Reference / Note: Automatic Reference Counting (ARC) is a compiler feature that manages object lifetimes automatically in Objective-C and Swift.

Question 4: A _____ is a build plan that defines the targets to build, the build configurations, and the test actions to execute.

a) **Scheme**
b) Target
c) File Owner
d) Group

Reference / Note: An Xcode scheme defines a collection of targets to build, a configuration to use when building, and a collection of tests to execute.

Question 5: The Organizer icon -> Documentation option in the Xcode Organizer window helps display _____.

a) Binary indexes for the files and symbols
b) Doc set
c) **Documentation Viewer**
d) Archives

Reference / Note: The Documentation option in Xcode Organizer opens the built-in Documentation Viewer.

Question 6: **Which of the primary elements does a storyboard use to design a user interface? (Choose all primary elements)**

a) **Dock, Scenes, and Views**

b) Controllers, Windows, and Pointers

c) Activities, Fragments, and Layouts

d) Canvas, Layers, and Assets

Reference / Note: In Xcode storyboards, the primary design components are Scenes (view controllers), Views, and the Dock (scene bar).

Question 7: **Identify the primary components of a version control tool integrated into Xcode.**

a) **Repository, Client, and Server**

b) Branch, Trunk, and Tag

c) Commit, Push, and Merge

d) Local, Staging, and Remote

Reference / Note: Version control architectures consist of the central repository, clients that clone and modify, and the server host.

Question 8: **Which of the following describes a tuple in Swift?**

a) A number that has a precision of at least 6 decimal digits

b) **A collection of zero or more values grouped into a single compound value**

c) A value that can be either true or false

d) Renderable text that is stored as an extended grapheme cluster

Reference / Note: Tuples group multiple values into a single compound value. The values within a tuple can be of any type.

Question 9: **Which of the following equations swaps the values of 'x' and 'y' in Swift using tuple assignment?**

a) **(x, y) = (y, x)**

b) y = x; x = y

c) x == y && y == x

d) swap(&x = &y)

Reference / Note: Swift allows simultaneous assignment through tuples: `(x, y) = (y, x)` cleanly swaps two variables without a temporary variable.

Question 10: **Which of the following describes the '||' operator in Swift?**

a) Logical AND operator

b) Logical NOT operator

c) **Logical OR operator**

d) Bitwise XOR operator

Reference / Note: The logical OR operator (a || b) is an infix operator that returns true if either expression is true.

Question 11: **Which of the following is an example of a closed range operator in Swift?**

a) **1...5**

b) 1..<5

c) 1..>5

d) 1?5

Reference / Note: The closed range operator (a...b) defines a range that runs from a to b, and includes both values a and b.

Question 12: What value does a bitwise AND operator return if 1 bit is true and one is false?

a) True (1)

b) **False (0)**

c) Undefined

d) Null

Reference / Note: A bitwise AND operator (&) returns 1 only if BOTH bits are 1; otherwise, it returns 0 (false).

Question 13: Which of the following statements explains the While loop in Swift?

a) **A loop in which the given condition is evaluated before execution of each iteration**

b) A set of code that executes a fixed number of times specified at start

c) A loop that always executes at least once before testing condition

d) A loop that never terminates

Reference / Note: A standard while loop evaluates its condition at the start of each pass through the loop.

Question 14: With which keyword does a function declaration start in Swift?

a) function

b) **func**

c) def

d) fn

Reference / Note: In Swift, functions are declared using the `func` keyword.

Question 15: How is an external argument label used with a function in Swift?

a) **When calling the function to make the call site expressive and sentence-like**

b) Inside the function body to reference local variables

c) Only when returning multiple values

d) To declare global variables across modules

Reference / Note: External argument labels are written before the parameter name and are used at the call site to clarify argument purpose.

Question 16: Which of the following will be executed if no matches occur in a Swift switch statement?

a) **default**

b) continue

c) fallthrough

d) break

Reference / Note: Swift switch statements must be exhaustive; the `default` case is executed when no other case pattern matches.

Question 17: Consider the following function: func addTwoInts(_ a: Int, _ b: Int) -> Int { return a + b }. What is the output of addTwoInts(2, 3)?

a) Result: 6

b) Result: 2

c) **Result: 5**

d) Result: 1

Reference / Note: addTwoInts(2, 3) executes 2 + 3 = 5.

Question 18: How can the snippet of code that assigns an enum case be rewritten using Swift type inference?

a) **student1 = .Art**

b) SubjectType = .Art

c) student1.SubjectType = .Art

d) SubjectType.student1 = .Art

Reference / Note: Once student1 is typed as SubjectType, its value can be reassigned with shorter dot syntax: `student1 = .Art`.

Question 19: How is a lazy stored property useful in Swift?

a) **When the initial value for the property is reliant on outside factors whose values are not known until after initialization**

b) It manipulates data very slowly to conserve CPU cycles

c) It stores values on disk instead of memory

d) It forces properties to remain constant throughout application lifetime

Reference / Note: A lazy stored property's initial value is not calculated until the first time it is accessed, ideal when dependent on outside factors.

Question 20: User interface files in an iOS project can have .nib or _____ as the filename extension.

a) **xib**

b) x

c) xb

d) xml

Reference / Note: Xcode Interface Builder uses .xib files (XML Interface Builder) for visual layouts, which are compiled into binary .nib files.

Question 21: What does a thread refer to in iOS application execution?

a) **A separate path of code execution within a process**

b) The program's console output stream

c) A persistent database connection

d) A network socket endpoint

Reference / Note: A thread represents an independent sequence of programmed instructions managed by the operating system scheduler.

Question 22: Which one of the following reduces the memory footprint of an iOS system by sharing common code?

a) **Framework**

b) Routine

c) Macro

d) Global variable

Reference / Note: Dynamic frameworks permit multiple applications to share code and resources in memory, reducing overall memory footprint.

Question 23: How many core collection types are provided by the Swift standard library?

a) **3 (Array, Set, Dictionary)**

b) 2 (List, Map)

c) 4 (Vector, List, Set, Table)

d) 1 (Array)

Reference / Note: Swift provides three primary collection types: Array (ordered), Set (unordered unique), and Dictionary (unordered key-value).

Question 24: _____ are typically used as a shortcut for accessing member elements of a collection, list, or sequence in Swift.

a) **Subscripts**
b) Methods
c) Protocols
d) Closures

Reference / Note: Subscripts enable accessing collection members via bracket syntax `[index]` without needing explicit lookup methods.

Question 25: The 'continue' statement instructs the compiler to _____ the current iteration of a loop and go to the next iteration.

a) **Stop execution of current loop body and proceed to next iteration**
b) Terminate the entire loop immediately
c) Restart the loop from iteration 0
d) Break out of all nested blocks

Reference / Note: In Swift, the continue statement stops the current iteration and jumps directly to the condition of the next iteration.

Question 26: Which of the following is the correct syntax for declaring a protocol in Swift?

a) **protocol SomeProtocol { // protocol definition goes here }**
b) protocol (SomeProtocol) { }
c) protocol = SomeProtocol { }
d) def protocol SomeProtocol { }

Reference / Note: Protocols in Swift are declared with `protocol ProtocolName { ... }`.

Question 27: In iOS development with SQLite, what does FMDB stand for?

a) **Flying Meat Database**
b) Function Monitoring Database
c) File Management Database
d) Fast Memory Data Block

Reference / Note: FMDB stands for Flying Meat Database, a popular Objective-C/Swift wrapper around SQLite created by Flying Meat Software.

Question 28: Which operator is used for downcasting a class instance to a subclass type when failure is possible?

a) **as?**
b) as!
c) is
d) cast<T>

Reference / Note: The conditional downcast operator `as?` returns an optional value of the target type (nil if downcast fails).

Question 29: Custom error types in Swift should conform to which standard protocol?

a) Error
b) **ErrorType (Swift 2) / Error (Swift 3+)**
c) Exception
d) Throwable

Reference / Note: In Swift, error handling models errors as values of types that conform to the `Error` protocol (formerly `ErrorType`).

Question 30: Which compiler flag is used in Xcode to disable Automatic Reference Counting for specific legacy Objective-C source files?

a) **-fno-objc-arc**

b) -fobjc-no-arc

c) -disable-arc

d) --no-arc

Reference / Note: The `-fno-objc-arc` compiler flag allows compiling individual legacy files without ARC in an otherwise ARC-enabled project.

Question 31: Which of the following panes can be viewed in the Debug Area in Xcode during an active debugging session?

a) **Debugger pane, Console pane, and Variable pane**

b) Navigator pane, Inspector pane, and Canvas pane

c) Timeline pane, Profiler pane, and Memory pane

d) Asset catalog pane, Schema pane, and Build pane

Reference / Note: The Debug Area in Xcode consists of the Variables Pane (displaying current variable states) and the Console Pane (debugger command prompt and output log).

Question 32: Which of the following must a developer have to download the latest software development kits, development tools, and example code from Apple?

a) Ad hoc provisioning profile

b) **Apple Developer Account**

c) Bundle Identifier

d) AppDelegate source files

Reference / Note: An Apple Developer Account is required to access official SDKs, beta toolsets, documentation archives, and sample source code on developer.apple.com.

Question 33: Which of the following statements are true regarding application resources in an iOS application package?

a) **They are static data files (images, audio, nibs) loaded by the application bundle at runtime**

b) They must be exclusively written and compiled into native C++ machine instructions

c) iOS does not support resources created using third-party design applications

d) They are executed directly by the Darwin operating system kernel at device boot

Reference / Note: Resources in iOS applications are static data files (such as .png, .json, .nib/.storyboard) stored in the application bundle and accessed at runtime.

Question 34: Identify the name of the icon that appears at the top of the Interface Builder canvas that handles the controller logic of the active scene view.

a) First Responder

b) Size Class Selector

c) Auto Layout Tools

d) **View Controller**

Reference / Note: In Xcode Interface Builder, the View Controller icon on the canvas top bar represents the controller instance managing the scene's view hierarchy and lifecycle.

Question 35: Which new files are automatically generated when a developer creates a project using Xcode's Single View Application template?

a) **App storyboard along with header (.h) and implementation (.m/.swift) files for the ViewController and AppDelegate classes**

b) Only the app storyboard file without any accompanying source code classes

c) Only an AppDelegate class without any UI view controller

d) Precompiled binaries of all UIKit view controller subclasses

Reference / Note: The Single View Application template creates a Main storyboard, an AppDelegate class, and a default ViewController implementation.

Question 36: When do mobile app developers typically choose to integrate an In-App Purchase (IAP) workflow into an iOS application?

a) **When the app offers premium digital content, virtual game items, subscriptions, or unlockable service features**

b) Exclusively when the app fails to generate initial profits from upfront App Store downloads

c) When the application is completely offline and requires no network connectivity

d) Only when paying annual Apple Developer Program maintenance fees

Reference / Note: In-App Purchases are used to sell digital goods, extra features, premium subscriptions, or virtual game items directly within the app.

Question 37: Swift provides first-class language error handling support for which of the following operations?

a) **Throwing, catching, and propagating recoverable errors at runtime**

b) Throwing and manipulating operating system Unix signals only

c) Catching low-level hardware segmentation faults automatically

d) Converting all runtime exceptions into silent null pointers

Reference / Note: Swift models error handling via `throws`, `do-catch`, and `try` statements to throw, catch, and propagate recoverable errors cleanly.

Question 38: If a Swift protocol requires a property to be both gettable and settable, that requirement CANNOT be fulfilled by which of the following?

a) **A constant stored property (`let`) or a read-only computed property (getter only)**

b) A variable stored property (`var`)

c) A read-write computed property with both getter and setter blocks

d) A stored property with property observers (`willSet` and `didSet`)

Reference / Note: A protocol requirement for a `{ get set }` property cannot be fulfilled by a constant (`let`) or read-only computed property because they cannot be modified.

Question 39: In Swift, initialization is the formal process of preparing an instance of which programming constructs for use?

a) **A Class, Structure, or Enumeration**

b) A Protocol and Extension only

c) A Global Function and Closure

d) A Tuple and Operator

Reference / Note: In Swift, initializers are called to create a new instance of a particular type (classes, structures, and enumerations).

Question 40: Which statement is used in Swift to transfer program control out of a scope if one or more conditions are not met (early exit)?

a) **guard**
b) if
c) unless
d) assert

Reference / Note: The Swift Programming Language Guide ('Control Flow'): A guard statement is used to transfer program control out of a scope on condition failure.

Question 41: In Swift, which language construct can add new functionality to an existing class, structure, enumeration, or protocol, but cannot override existing functionality?

a) **Extensions**
b) Generics
c) Subclasses
d) Protocols

Reference / Note: The Swift Programming Language Guide ('Extensions'): Extensions can add new functionality to a type, but they cannot override existing functionality.

Question 42: In Swift, which range operators are available to express a range of values?

a) **Closed range operator (...) and Half-open range operator (..<)**
b) Inclusive operator (->) and Exclusive operator (<-)
c) Delta range (..) and Infinity range (...)
d) Open range (::) and Bounded range (..)

Reference / Note: The Swift Programming Language Guide ('Basic Operators'): Swift includes two range operators: the closed range operator (a...b) and the half-open range operator (a..<b).

Question 43: How many kinds of Types does the Swift programming language support, and what are they?

a) **2 kinds: Named types and Compound types**
b) 3 kinds: Value types, Reference types, and Dynamic types
c) 4 kinds: Class types, Struct types, Enum types, and Protocol types
d) 1 kind: Universal Any type

Reference / Note: The Swift Programming Language Guide ('Types'): In Swift, there are two kinds of types: named types (classes, structures, enumerations, and protocols) and compound types (functions and tuples).

Question 44: Which keyword in Swift is used to define an alternative name (alias) for an existing data type?

a) **typealias**
b) typedef
c) alias
d) retype

Reference / Note: The Swift Programming Language Guide ('The Basics'): Type aliases define an alternative name for an existing type using the typealias keyword.